

Axomeme Architecture Specification

A Self-Contained Mathematical and Algorithmic Specification for Neural Detection of Natural Selection in Molecular Sequences

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Abstract

This document provides the authoritative mathematical and algorithmic specification of the **Axomeme** neural network architecture. Axomeme is an end-to-end deep learning framework designed to predict site-level natural selection (Likelihood Ratio Test statistics) directly from multi-species codon sequence alignments and phylogenetic tree matrices. The architecture combines a 4-layer dual-axial transformer engine with an evolutionarily grounded **Markov Exponential Decay Kernel** ($P(t) = e^{-\lambda t}$), 4D metric tree Multidimensional Scaling (MDS), block-diagonal track disentanglement between synonymous and non-synonymous features, a **12-Expert Mixture-of-Pooling Streams (MoPS)** gating network, and a rank-consistent CORAL ordinal prediction head.

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1 Introduction & Problem Formulation

1.1 The Statistical Framework of Natural Selection

In evolutionary biology, protein-coding gene sequences evolve under varying selective pressures:

1. **Synonymous Substitutions (α or dS):** Nucleotide changes that do not alter the translated amino acid.
2. **Non-Synonymous Substitutions (β or dN):** Nucleotide changes that alter the translated amino acid.

Site-level selection is formulated as a hypothesis test:

$$H_0 \text{ (Null Hypothesis) : } \beta \leq \alpha \quad (\text{Neutral or Purifying Selection}) \quad (1)$$

$$H_1 \text{ (Alternative Hypothesis) : } \beta^+ > \alpha \quad (\text{Episodic Positive Selection on a subset of branches}) \quad (2)$$

The Likelihood Ratio Test (LRT) statistic is defined as:

$$\text{LRT} = 2 \cdot \left(\log \mathcal{L}(H_1) - \log \mathcal{L}(H_0) \right) \quad (3)$$

Axomeme models the mapping from raw multi-species sequence alignments and tree matrices directly to LRT statistics, replacing numerical likelihood optimization with a direct neural operator.

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2 Input Representation & Encoding

Consider an alignment matrix X containing N species sequences across S codon sites.

2.1 Discrete Sequence Tokenization

The alignment is tokenized into two discrete integer matrices:

1. **Codon Tokens ($X_{\text{codon}} \in \{0, \dots, 65\}^{N \times S}$):** 64 sense codons + 2 termination/gap tokens.
2. **Amino Acid Tokens ($X_{\text{aa}} \in \{0, \dots, 22\}^{N \times S}$):** 20 standard amino acids (indices 0–19) + 1 stop codon (20) + 1 gap token (21) + 1 unknown token (22).

2.2 Phylogenetic Distance & Coordinates

1. **Pairwise Patristic Distance Matrix ($\mathbf{T}_{\text{phylo}} \in \mathbb{R}^{N \times N}$):** Patristic branch distances are computed from the unrooted phylogenetic tree:

$$\tau_{ij} = \sum_{b \in \text{Path}(i,j)} \text{Length}(b) \quad (4)$$

2. **Pairwise Jukes-Cantor Distance Fallback:** If branch lengths are absent, pairwise distances are calculated directly from sequence identity:

$$d(s_i, s_j) = -\frac{3}{4} \ln \left(1 - \frac{4}{3} p_{ij} \right) \quad (5)$$

3. **4D Multidimensional Scaling (MDS) Coordinates ($M \in \mathbb{R}^{N \times 4}$):** The distance matrix $\mathbf{T}_{\text{phylo}}$ is projected into 4-dimensional Euclidean spatial coordinates via classical metric MDS.

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3 Block-Diagonal Track Disentanglement

To isolate non-synonymous amino acid selection signals from synonymous nucleotide noise, Axomeme enforces **Track Disentanglement** throughout the embedding and transformation layers.

3.1 Vector Partitioning

Embedding vectors $\mathbf{x} \in \mathbb{R}^{256}$ are partitioned into two independent 128-dimensional sub-vectors:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_{\text{codon}} \\ \mathbf{x}_{\text{aa}} \end{bmatrix} \in \mathbb{R}^{256}, \quad \mathbf{x}_{\text{codon}} \in \mathbb{R}^{128}, \quad \mathbf{x}_{\text{aa}} \in \mathbb{R}^{128} \quad (6)$$

3.2 The BlockLinear Operator

All linear projection layers use a block-diagonal weight matrix **BlockLinear**:

$$\mathbf{W}_{\text{block}} = \begin{pmatrix} \mathbf{W}_{\text{codon}} & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_{\text{aa}} \end{pmatrix} \in \mathbb{R}^{256 \times 256} \quad (7)$$

$$\mathbf{W}_{\text{block}}\mathbf{x} = \begin{bmatrix} \mathbf{W}_{\text{codon}}\mathbf{x}_{\text{codon}} \\ \mathbf{W}_{\text{aa}}\mathbf{x}_{\text{aa}} \end{bmatrix} \quad (8)$$

4 The Dual-Axial Attention Engine & Markov Exponential Decay Kernel

4.1 Species Row Attention & Markov Distance Kernel

For each attention head $h \in \{1, \dots, 8\}$:

1. Raw Query-Key Similarity Score:

$$S_{ij}^{(h)} = \frac{(\mathbf{q}_i^{(h)})^\top \mathbf{k}_j^{(h)}}{\sqrt{32}} \quad (9)$$

2. **Markov Continuous-Time Decay Kernel** ($\tilde{S}_{ij}^{(h)} \in \mathbb{R}$): Under continuous-time Markov substitution models ($P(t) = e^{\mathbf{Q}t}$), evolutionary influence between species separated by patristic distance τ_{ij} decays exponentially: $K(\tau_{ij}) \propto e^{-\lambda_h \cdot \tau_{ij}}$. In attention score logit space, this corresponds to subtracting a learnable positive distance penalty:

$$\tilde{S}_{ij}^{(h)} = \frac{(\mathbf{q}_i^{(h)})^\top \mathbf{k}_j^{(h)}}{\sqrt{32}} - \lambda_h \cdot \tau_{ij}, \quad \text{where } \lambda_h = \text{Softplus}(w_{\text{phylo},h}) > 0 \quad (10)$$

3. Softmax Normalization:

$$A_{ij}^{(h)} = \frac{\exp(\tilde{S}_{ij}^{(h)})}{\sum_{m=1}^N \exp(\tilde{S}_{im}^{(h)})} \quad (11)$$

5 Mixture-of-Pooling Streams (MoPS) Gating Architecture

Following 4 layers of phylogenetic species attention, each species i at site s has a 256-dimensional embedding vector $\mathbf{h}_i \in \mathbb{R}^{256}$. Axomeme pools species representations across $K = 12$ specialist experts:

Table 1: The 12 MoPS Specialist Pooling Experts

Expert Stream	Dims	Mathematical Definition	Primary Biological Function
1. mean_pooled	256	$\boldsymbol{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i$	Equilibrium Consensus Background
2. max_pooled	256	$\mathbf{m} = \max_i \mathbf{h}_i$	Deep Mutant Outlier Burst
3. min_pooled	256	$\mathbf{n} = \min_i \mathbf{h}_i$	Purifying Constraint Floor
4. cat_pooled	256	$\text{MLP}_{\text{fusion}} \left(\left[\text{MLP}_{\text{stream}} \left(\mathbf{e}_a \parallel p_a \right) \right]_{a=0}^{19} \right)$	Clean 20-Class Allele Matrix
5. diff_pooled	256	$\text{ReLU}(\mathbf{m} - \boldsymbol{\mu}) \in \mathbb{R}^{256}$	Full 256D Codon + AA Disentangled Range
6. var_pooled	256	$\frac{1}{N} \sum_{i=1}^N (\mathbf{h}_i - \boldsymbol{\mu})^2 \in \mathbb{R}^{256}$	2nd Moment (Feature Variance)
7. skew_pooled	256	$\frac{1}{N} \sum_{i=1}^N (\mathbf{h}_i - \boldsymbol{\mu})^3 \in \mathbb{R}^{256}$	3rd Moment (Asymmetric Lineage Acceleration)
8. kurt_pooled	256	$\frac{1}{N} \sum_{i=1}^N (\mathbf{h}_i - \boldsymbol{\mu})^4 \in \mathbb{R}^{256}$	4th Moment (Rare Outlier Tailness)
9. lse_pooled	256	$\log \left(\frac{1}{N} \sum_{i=1}^N \exp(\mathbf{h}_i) \right)$	Smooth Log-Sum-Exp Max/Mean Interpolation
10. phylo_mean_pooled	256	$\frac{\sum_{i=1}^N w_i \mathbf{h}_i}{\sum_{i=1}^N w_i}, \quad w_i = \frac{1}{d_i + 1.0}$	Tree-Weighted Consensus
11. dnds_contrast	256	$\frac{1}{N} \sum_{i=1}^N \left(\frac{\ \mathbf{h}_{i,\text{aa}}\ }{\ \mathbf{h}_{i,\text{codon}}\ + \epsilon} \right) \mathbf{h}_i$	Non-synonymous vs Synonymous Channel Norm Ratio
12. phylo_var_pooled	256	$\frac{\sum_{i=1}^N w_i (\mathbf{h}_i - \boldsymbol{\mu}_{\text{phylo}})^2}{\sum_{i=1}^N w_i}$	Tree-Weighted Lineage Variance

5.1 Dynamic Neural Routing

The gating network computes Softmax routing weights over the 12 expert streams based on site consensus:

$$\mathbf{g} = \text{Softmax}(\mathbf{W}_g \boldsymbol{\mu} + \mathbf{b}_g) \in [0, 1]^{12}, \quad \sum_{k=1}^{12} g_k = 1.0 \quad (12)$$

$$\mathbf{h}_{\text{pooled}} = \text{stream_fusion} \left(\sum_{k=1}^{12} g_k \cdot \mathbf{s}_k \right) \in \mathbb{R}^{256} \quad (13)$$

6 Rank-Consistent CORAL Ordinal Head

The pooled site vector $\mathbf{h}_{\text{pooled}} \in \mathbb{R}^{256}$ is passed to the Consistent Rank Logits (CORAL) ordinal prediction head.

6.1 Non-Linear Projection & Strict Logit Monotonicity

To prevent 1D feature collapse, features are projected through a 2-layer MLP projection $\text{proj}(\mathbf{h}_{\text{pooled}})$ before adding monotonic threshold cutoffs b_k :

$$\text{proj}(\mathbf{h}_{\text{pooled}}) = \mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{h}_{\text{pooled}}), \quad \mathbf{W}_1 \in \mathbb{R}^{64 \times 256}, \quad \mathbf{W}_2 \in \mathbb{R}^{1 \times 64} \quad (14)$$

Threshold cutoffs b_k enforce strict probability monotonicity ($P(y > b_0) \geq P(y > b_1) \geq \dots$):

$$b_0 = \text{Learnable Parameter}, \quad b_k = b_0 - \sum_{j=1}^k \text{Softplus}(\theta_j) \quad (15)$$

The cumulative logits for ordinal threshold $k \in \{0, \dots, 11\}$ are:

$$z_k(\mathbf{h}_{\text{pooled}}) = \text{proj}(\mathbf{h}_{\text{pooled}}) + b_k \quad (16)$$

6.2 Expected Value Decoding

Continuous LRT predictions are decoded directly via Smooth Expected Value Interpolation over the empirical MEME χ^2 mixture threshold midpoints:

$$\text{LRT} = \sum_{m=0}^{12} P(\text{Bin } m) \cdot \text{Midpoint}_m \quad (17)$$

where Midpoint_m represents the raw LRT midpoint of bin m , providing continuous, un-biased selection strength estimates.